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Instantaneous Tunneling Flight Time for Wavepacket Transmission Through Asymmetric Barriers

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Abstract

The time it takes a particle to tunnel through the asymmetric Eckart barrier potential is investigated using Gaussian wavepackets, where the barrier serves as a model for the potential along a chemical reaction coordinate. We have previously shown that the, in principle experimentally measurable, tunneling flight time, which determines the time taken by the transmitted particle to traverse the barrier, vanishes for symmetric potentials like the Eckart and square barrier.[Petersen, J.; Pollak, E. *J. Phys. Chem. Lett.* **2017**, 9, 4017.] Here we show that the same result is obtained for the asymmetric Eckart barrier potential, and therefore, the zero tunneling flight time seems to be a general result for one-dimensional time-independent potentials. The wavepacket dynamics is simulated using both an exact quantum mechanical method and a classical Wigner prescription. The excellent agreement between the two methods shows that quantum coherences are not important in pure one-dimensional tunneling and reinforces the conclusion that the tunneling flight time vanishes.

1 Introduction

The question of how much time it takes a particle to tunnel through a potential barrier has intrigued the community for over ninety years. The non-classical phenomenon of tunneling was predicted shortly after the advent of quantum mechanics in the 1920's.¹ For a long time though there was a lack of consensus over a definitive answer to the tunneling time question. This was partly due to the various definitions of time in quantum mechanics^{2,3} and the different results for the tunneling time that they produced.⁴⁻⁸ Lately, attosecond strong field ionization experiments⁹ have addressed the tunneling time enigma but also without giving a definite conclusion.^{10,11}

We have recently used the formalism of transition path time distributions¹² to study this question. In its various formulations, the transition path time distribution gives the probability distribution of transition times between two spatial points. It has been successfully applied, both theoretically and experimentally, to molecular processes behaving classically, e.g., protein folding.¹³⁻¹⁶ A quantum mechanical counterpart of the transition path time distribution is expressed in terms of time correlation functions.¹⁷⁻¹⁹ Such a time distribution for spatial transitions in quantum systems may in principle be experimentally measurable, e.g., in single atom time-of-flight experiments,²⁰ where particles are released from a trap²¹ at time zero, and the arrival time at a detector screen is recorded.

Using the transition path time distribution, we formulated recently a tunneling flight time²² by comparing the mean flight time of transmitted and reflected particles scattered through a symmetric barrier. For an incident Gaussian wavepacket with a range of incident momenta, determined by a width parameter Γ , one finds that the transmission through the barrier preferentially filters the higher momentum components so that the mean transmission time is faster than the mean reflection time. This filtering effect masks the tunneling time but becomes smaller as the momentum width $\hbar^2\Gamma$ becomes smaller. The tunneling flight time is then observed by considering the flight time difference in the limit that $\hbar^2\Gamma \rightarrow 0$. For symmetric barriers we showed that this flight time vanishes.

Symmetric exchange is but a small subset of possible reactions. In this paper we study the more chemically relevant and more general setup of tunneling through an asymmetric potential energy profile along a one-dimensional reaction coordinate. Without loss of generality we will consider exothermic reactions, that is the potential energy in the asymptotic region to the right of the barrier is lower than that to the left. Due to this asymmetry, the motion of transmitted particles in the asymptotic region, which initially are incident on the barrier from the left will be faster than the motion of the reflected particles but this time difference has nothing to do with the tunneling time itself. To elucidate the tunneling flight time in the asymmetric case it is therefore necessary to consider the motion of the scattered particle in the forward and backward directions. The central theme of this paper is to study the tunneling flight time in this asymmetric case.

The dynamics of the particle is described by the time evolution of a Gaussian wavepacket which is propagated using both exact quantum mechanics and a classical Wigner approximation to the exact result. The latter is constructed such that the tunneling flight time vanishes explicitly. The two propagation methods lead to the identical conclusion which is that also in the asymmetric case, the tunneling flight time vanishes. The comparison also suggests that quantum interference effects are not important and that in this sense tunneling through the asymmetric barrier may also be considered as an incoherent process.²⁴

The theoretical framework is presented in Section 2; the tunneling flight time for the asymmetric case is defined, the asymmetric Eckart potential model to be studied is described, and the propagation methods to be used are presented. Numerical results are then given in Section 3. Finally, we present some concluding remarks in Section 4.

2 Theory

2.1 Transition path time distribution

We consider a particle with mass M scattered by a potential $V(x)$ whose Hamiltonian is given by

$$\hat{H} = \frac{\hat{p}^2}{2M} + V(\hat{x}). \quad (2.1)$$

$\hat{x}$ and $\hat{p} = -i\hbar\partial/\partial\hat{x}$ are the position and momentum operators, respectively. The particle's statistics are given by an incident coherent state centered about the phase space point (p_i, x_i) :

$$\langle x|\Psi\rangle = \left(\frac{\Gamma}{\pi}\right)^{1/4} \exp\left(-\frac{\Gamma}{2}(x-x_i)^2 + \frac{i}{\hbar}p_i(x-x_i)\right). \quad (2.2)$$

The width parameter Γ controls the uncertainty in the initial position and momentum; as $\hbar^2\Gamma \rightarrow 0$, the incident momentum is increasingly well defined while the spatial width of the wavepacket grows indefinitely.

The transition path time distribution for scattering of the initial wavepacket is defined in terms of the positive density correlation function at time t about the final position y

$$C_t(y; \Psi) = \text{Tr} \left[|\Psi\rangle\langle\Psi| \hat{K}_t^\dagger \delta(\hat{x} - y) \hat{K}_t \right] = |\langle y|\hat{K}_t|\Psi\rangle|^2, \quad (2.3)$$

where $\hat{K}_t = \exp(-i\hat{H}t/\hbar)$ is the time evolution operator.²² The transition path time probability distribution is

$$P_t(y; \Psi) = \frac{C_t(y; \Psi)}{\int_0^\infty dt C_t(y; \Psi)}. \quad (2.4)$$

In Appendix A, we show that due to the quantum phenomenon of threshold quantum reflection, at long time, the transition path time distribution vanishes as t^{-3} ,²³ so that the normalization integral and the mean time of the distribution are well defined.

The transition path time probability distribution is partitioned into two parts, one for

the reflected and one for the transmitted wavepacket, i.e.,

$$P_t(y; \Psi) = |R_\Psi|^2 P_{R,t}(-y; \Psi) + |T_\Psi|^2 P_{T,t}(y; \Psi), \quad (2.5)$$

where $|R_\Psi|^2$ and $|T_\Psi|^2$ are the averaged reflection and transmission probabilities

$$|R_\Psi|^2 = \int_{-\infty}^{\infty} dk \frac{|R(k)|^2}{\sqrt{\pi\Gamma}} \exp\left(-\frac{(\hbar k - p_i)^2}{\hbar^2 \Gamma}\right) \quad (2.6)$$

$$|T_\Psi|^2 = \int_{-\infty}^{\infty} dk \frac{|T(k)|^2}{\sqrt{\pi\Gamma}} \exp\left(-\frac{(\hbar k - p_i)^2}{\hbar^2 \Gamma}\right). \quad (2.7)$$

Here $R(k)$ and $T(k)$ are the usual reflection and transmission amplitudes, respectively, for the potential in question. The expression in Eq. (2.5) holds for the particle incident from the left ($x_i < 0$), but in the case of the particle approaching the barrier from the right, the sign of y is changed on the right hand side of Eq. (2.5).

To study only the tunneling process and eliminate unwanted trivial effects, some restrictions on the wavepacket parameters are necessary. The incident energy $E = p_i^2/(2M) + V(x_i)$ has to be smaller than the maximum of the potential $V_{\max}$. Tunneling implies that all particles described by the incident wavepacket are transmitted via tunneling, rather than above barrier transmission. This implies that the variance of the wavepacket energy has to be sufficiently smaller than $V_{\max} - E$ such that the probability for having an initial energy greater than the barrier height is much smaller than the transmission probability at the incident energy E . Similarly, the incident wavepacket must be located sufficiently far away from the barrier so that the initial probability of finding the particle in the barrier region is also much smaller than the transmission probability. In other words, $|x_i|$ should be much larger than the spatial width of the initial wavepacket ($1/\sqrt{\Gamma}$). To separate between the incident wavepacket and its reflected portion, the final point ($y > 0$), at which the distributions are observed, is chosen such that $y - |x_i| \gg 1/\sqrt{\Gamma}$. With this choice the amplitude of the initial wavepacket at $\pm y$ is negligible.

For the particle incident from the left, the mean flight time $\langle t \rangle$ for arrival at $x = -y$ (reflection) and $x = y$ (transmission) is defined as

$$\langle t \rangle_{R/T,l} = \int_0^\infty dt t P_{R/T,t}(\mp y; \Psi_l). \quad (2.8)$$

Similarly one defines the mean flight times for the particle incident from the right $\langle t \rangle_{R/T,r}$ except that now the sign of y is inverted. We then consider separately the flight time difference for particles incident upon the barrier from the left $\Delta t_l(\Gamma)$ and from the right $\Delta t_r(\Gamma)$ which is defined as:

$$\Delta t_{l/r}(\Gamma) = \langle t \rangle_{R,l/r} - \langle t \rangle_{T,l/r} \quad (2.9)$$

Due to the asymmetry, it is necessary to eliminate flight time differences caused by the different asymptotic potential energies. Therefore we consider the overall flight time difference defined as the average²²

$$\overline{\Delta t}(\Gamma) = \frac{1}{2} [\Delta t_l(\Gamma) + \Delta t_r(\Gamma)]. \quad (2.10)$$

Given the incident coherent state wavepacket in Eq. (2.2), the probability of tunneling on the incident momentum would go as $\exp \left[-\frac{(p-p_i)^2}{\hbar^2 \Gamma} + \ln(|T(p/\hbar)|^2) \right]$. This means that the most probable momentum, obtained by minimizing the exponent and using the condition $\hbar\sqrt{\Gamma} \ll p_i$, becomes

$$p_{\max} \simeq p_i + \frac{\hbar^2 \Gamma}{2} \left[\frac{d}{dp} \ln(|T(p/\hbar)|^2) \right]_{p=p_i}. \quad (2.11)$$

In the tunneling regime, the transmission coefficient is a monotonically increasing function of the energy so that the second term on the right hand side is positive. The most probable momentum for the transmitted part is thus larger than the incident momentum demonstrating the filtering effect and its linear dependence on the width parameter Γ . This filtering effect masks any time scale that has to do with the tunneling time, and therefore one would want to remove it. For this purpose we study the limiting behaviour of $\overline{\Delta t}(\Gamma)$ when $\Gamma \rightarrow 0^+$.

This limit is then defined to be the tunneling flight time (TFT)²²

$$t_{\text{TFT}} \equiv \lim_{\Gamma \rightarrow 0^+} [-\overline{\Delta t}(\Gamma)]. \quad (2.12)$$

In practice, since the spatial width of the wavepacket is approaching infinity in this limit, one has to increase $|x_i|$ and y as the width parameter Γ is decreased, such that the tunneling conditions considered above are not violated.

2.2 Asymmetric Eckart potential

In this paper, the tunneling flight time will be studied for the case of an asymmetric Eckart barrier potential²⁵

$$V(x) = V_{-\infty} \left(\frac{1}{1 + \exp(x/b)} + \frac{\kappa \exp(x/b)}{[1 + \exp(x/b)]^2} \right), \quad (2.13)$$

for which the asymptotic behaviours are $\lim_{x \rightarrow -\infty} V(x) = V_{-\infty}$ and $\lim_{x \rightarrow \infty} V(x) = 0$. The parameter κ determines the shape of the potential, i.e., $\kappa \in [-1, 1]$ leads to a smooth step potential, while other values give a barrier or a well potential. We will use the values $V_{-\infty} = 1$ and $b = 1$ (in atomic units) such that a barrier potential is obtained for $\kappa > 1$. The resulting potential is plotted in panel a) of Fig. 1 for the values $\kappa = 4, 5, 6, 7, 8$.

The reflection and transmission amplitudes at an incident energy E are given by²⁵

$$R(k) = \frac{\Gamma\left(\frac{1}{2} + \omega - ibk - ibk'\right) \Gamma\left(\frac{1}{2} - \omega - ibk - ibk'\right) \Gamma(2ibk')}{\Gamma\left(\frac{1}{2} + \omega - ibk + ibk'\right) \Gamma\left(\frac{1}{2} - \omega - ibk + ibk'\right) \Gamma(-2ibk')} \quad (2.14)$$

$$T(k) = \frac{\Gamma\left(\frac{1}{2} + \omega - ibk - ibk'\right) \Gamma\left(\frac{1}{2} - \omega - ibk - ibk'\right)}{\Gamma(1 - 2ibk) \Gamma(-2ibk')} \sqrt{\frac{k}{k'}}, \quad (2.15)$$

where $\hbar k = \sqrt{2ME}$, $\hbar k' = \sqrt{2M(E - V_{-\infty})}$, and $4\omega^2 = 1 - 8MV_{-\infty}b^2\kappa/\hbar^2$. The position

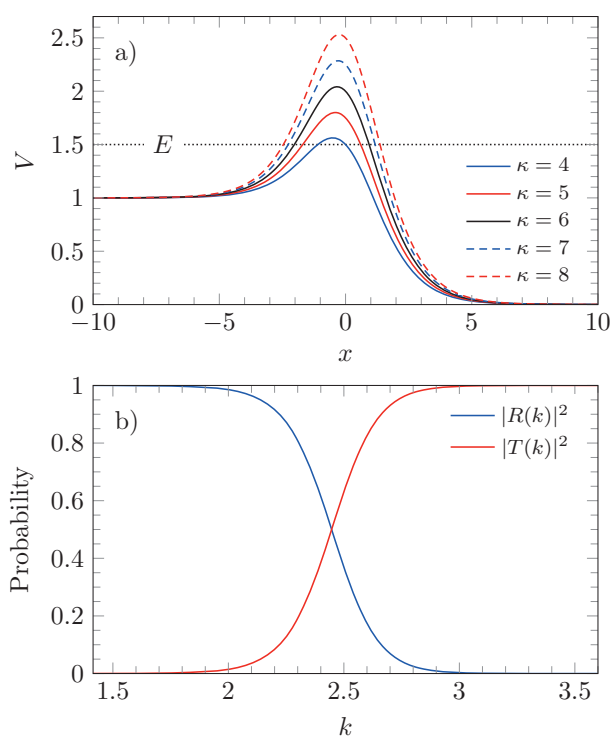


Figure 1: The asymmetric Eckart barrier with $V_{-\infty} = 1$ and $b = 1$. In panel a) the barrier potential is plotted as a function of the distance x for five values of the asymmetry parameter κ . The incident energy $E = 3/2$ is also shown. In panel b) the reflection and transmission probabilities are plotted as a function of the incident momentum $\hbar k$ for the case of $\kappa = 4$. All dimensional variables are given in atomic units.

representation of the scattering eigenfunction is

$$\langle x|k^+\rangle = \sqrt{\frac{k'}{k} \frac{T(k)}{\sqrt{2\pi}}} \tilde{x}^{ibk'} (1 - \tilde{x})^{-ibk} \times F\left(\frac{1}{2} + \omega - ibk + ibk', \frac{1}{2} - \omega - ibk + ibk'; 1 - 2ibk, 1 - \tilde{x}\right), \quad (2.16)$$

where we used the auxiliary variable $\tilde{x} = [1 + \exp(-x/b)]^{-1}$ and F denotes Gauss' hypergeometric function).²⁵ The asymptotic forms of Eq. (2.16) are

$$\langle x|k^+\rangle = \frac{1}{\sqrt{2\pi}} [\exp(ik'x) + R(k') \exp(-ik'x)], x \rightarrow -\infty, k' > 0 \quad (2.17)$$

$$\langle x|k^+\rangle = \sqrt{\frac{k'}{k} \frac{T(k')}{\sqrt{2\pi}}} \exp(ikx), x \rightarrow \infty, k' > 0 \quad (2.18)$$

$$\langle x|k^+\rangle = \frac{1}{\sqrt{2\pi}} [\exp(ikx) + R^*(k) \exp(-ikx)], x \rightarrow \infty, k' < 0 \quad (2.19)$$

$$\langle x|k^+\rangle = \sqrt{\frac{k'}{k} \frac{T^*(k)}{\sqrt{2\pi}}} \exp(ik'x), x \rightarrow -\infty, k' < 0. \quad (2.20)$$

2.3 Quantum mechanical propagation

The quantum mechanical time evolution of the initial wavepacket is carried out using the complete set of scattering states in Eq. (2.16) such that

$$\langle x|\hat{K}_t|\Psi\rangle = \int_{-\infty}^{\infty} dk \exp\left(-i\frac{\hbar k^2}{2M}t\right) \langle x|k^+\rangle \langle k^+|\Psi\rangle. \quad (2.21)$$

Since the initial wavepacket is localized in the asymptotic region, the matrix element $\langle k^+|\Psi\rangle$ can be evaluated analytically using the asymptotic forms of $\langle x|k^+\rangle$ in Eqs. (2.17)–(2.20), e.g., in the case that the initial wavepacket is localized on the left hand side of the barrier, the matrix element reads

$$\langle k^+|\Psi\rangle = \left(\frac{1}{\pi\Gamma}\right)^{1/4} \left[\exp\left(-\frac{(\hbar k' - p_i)^2}{2\hbar^2\Gamma} - ik'x_i\right) + R^*(k') \exp\left(-\frac{(\hbar k' + p_i)^2}{2\hbar^2\Gamma} + ik'x_i\right) \right]. \quad (2.22)$$

In the numerical calculations, the quadrature over k in Eq. (2.21) is carried out in the range $[p_i/\hbar - 7\sqrt{\Gamma}; p_i/\hbar + 7\sqrt{\Gamma}]$ using 10^5 equidistant grid points.

2.4 Classical Wigner propagation

The wavepacket dynamics is also evaluated by a classical Wigner propagation method, as introduced in Ref. 22. The density correlation function C_t in Eq. (2.3) may be rewritten as a phase space trace of two Wigner densities; one is the Wigner representation of the initial wavepacket

$$\rho_{\Psi}(x, p) = \frac{1}{\pi\hbar} \exp \left(-\Gamma (x - x_i)^2 - \frac{(p - p_i)^2}{\hbar^2\Gamma} \right), \quad (2.23)$$

and the other one is the Wigner representation of the Heisenberg time evolved density $\hat{\rho}(t) = \hat{K}_t^\dagger \delta(\hat{x} - y) \hat{K}_t$. In the classical Wigner approximation, the latter is approximated by its classical counterpart $\delta(x_t - y)$, where x_t is the final position of the classical trajectory with initial conditions (x, p) evolving for time t .

In order to study tunneling we add a non-classical feature to the trajectory dynamics; when a trajectory with incident momentum p reaches a classical turning point, a “reflected” trajectory is created with probability $|R(p)|^2$ and a “transmitted” trajectory is instantaneously created at the other turning point of the barrier potential with probability $|T(p)|^2$. *Hence, the tunneling time of each trajectory is explicitly assumed to be zero.* With this prescription, the classical Wigner approximation includes the momentum filtering effect, since the trajectories carry probabilities obtained from the reflection and transmission amplitudes in Eqs. (2.14) and (2.15). This classical Wigner theory provides an incoherent description of the tunneling dynamics. In the subsequent numerical calculations, the Wigner phase space integral is evaluated stochastically using Monte-Carlo with $\sim 10^{10}$ classical trajectories.

3 Results and discussion

Atomic units (e.g., $\hbar = 1$) are used throughout this paper. The mass $M = 1$ and the barrier parameters $V_{-\infty} = 1$ and $b = 1$ are used in all numerical calculations. In Fig. 2 we show an example of the space and time dependent quantum probability density $|\langle x | \hat{K}_t | \Psi \rangle|^2$ of an initial coherent state wavepacket moving on the asymmetric Eckart barrier potential with $\kappa = 4$. In panel a) the incident wavepacket starts out to the left of the barrier ($x_i = -50$) with a center momentum $p_i = 1$ and the width $\Gamma = 10^{-2}$. In this case the wavepacket is transmitted through the barrier with a probability of 0.428. In panel b) the incident wavepacket is positioned on the right hand side of the barrier ($x_i = 50$) with $p_i = -\sqrt{3}$ and $\Gamma = 10^{-2}$, and the transmission probability is 0.434. The incident energy is the same for the two wavepackets with $E = p_i^2/(2M) + V(x_i) = 3/2$.

This example is chosen to illustrate the setup with two independent wavepacket propagations; one wavepacket is approaching the barrier potential from the left and the other one is approaching from the right. One of the necessary conditions for tunneling, i.e., the variance of the wavepacket energy is much smaller than $V_{\max} - E$, is not satisfied in this example. In fact 19 % of the wavepacket's energy components are above the barrier height $V_{\max} = 1.5625$, but the transmitted wavepackets are difficult to visualize if tunneling conditions are met. Furthermore, the qualitative behavior of the scattering processes in time and space is not significantly dependent on whether the tunneling conditions are fully met or not.

In Fig. 2a) we notice that the transmitted wavepacket moves faster than the reflected one, which moves at approximately the same speed as the incident wavepacket. This is expected due to the form of the potential, see panel c), for which the initial momentum $p_i = \sqrt{2M(E - V_{-\infty})}$, $E > V_{\max}$, of a classical particle traversing the barrier from the left is smaller than the final momentum $p_f = \sqrt{2ME}$. During the scattering event, an interference pattern is observed on the left hand side of the barrier, where incoming and reflected wave components interfere. Not surprisingly the opposite scenario holds in panel b), where the transmitted wavepacket moves slower than the reflected one (which moves at

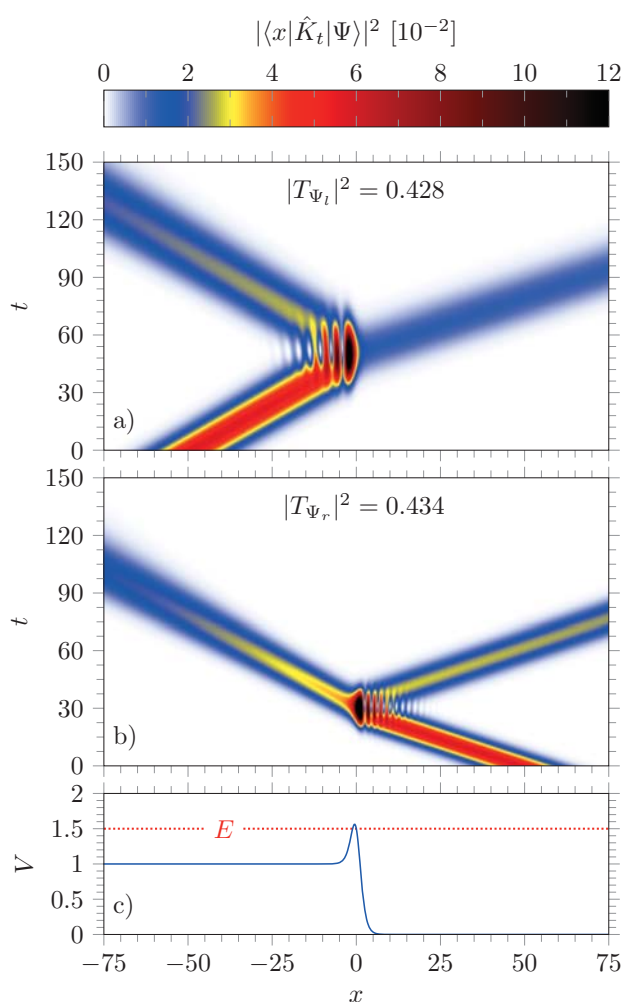


Figure 2: The space and time evolution of a Gaussian wavepacket scattered from an asymmetric Eckart barrier with the parameters $V_{-\infty} = 1$, $b = 1$, and $\kappa = 4$ [as plotted in panel c)]. In panel a) the incident wavepacket is positioned in the asymptotic region to the left of the barrier at $x_i = -50$ and the center momentum is $p_i = 1$. In panel b) the wavepacket is initially positioned on the right hand side of the barrier, and the phase space center is $(p_i, x_i) = (-\sqrt{3}, 50)$. In both cases the width of the initial wavepacket is $\Gamma = 10^{-2}$ and the incident energy is $E = p_i^2/(2M) + V(x_i) = 3/2$.

approximately the same speed as the incident wavepacket), and the interference pattern is observed on the right hand side of the barrier. After the scattering events, the reflected and transmitted wavepackets have approximate Gaussian shapes. We distinguish between the two independent wavepacket propagations by labeling the initial wavepacket with a subscript, i.e., $\Psi_{l/r}$ is the initial wavepacket approaching the barrier potential from the left/right.

From inspection of Fig. 2, transition path time probability distributions calculated at $\pm y = \pm 75$ are shifted with respect to each other, e.g., $P_{R,t}(-y; \Psi_l)$ attains its maximum at later times than $P_{T,t}(+y; \Psi_l)$ [see Fig. 2a)]. In other words, and as mentioned above, the transmitted wavepacket moves faster than the reflected one, when the incident wavepacket approaches the barrier from the left, and vice versa, when the incident wavepacket comes from the right. This is also evident from panel a) in Fig. 3, where the four time probability distributions at $\pm y = \pm 75$ are shown. If the wavepacket parameters are chosen such that all tunneling conditions in Section 2.1 are fulfilled, the transition path time probability distributions can be fully separated, see Fig. 3b). Here the center position of the incident wavepackets is $|x_i| = 2 \cdot 10^5$, the center momenta are the same as before, the width is $\Gamma = 5 \cdot 10^{-8}$, the potential parameter κ is 6, and the time distributions are detected at $\pm y = \pm 4 \cdot 10^5$.

To elucidate the tunneling flight time it is necessary to consider the overall flight time difference $\overline{\Delta t}$ and study its behaviour in the limit that the width parameter Γ approaches zero so that the momentum filtering effect of the barrier is eliminated, and therefore, the flight time differences are not affected by the different momentum distributions of the reflected and transmitted wavepackets. In Fig. 4 we plot $\overline{\Delta t}$ as a function of Γ for $\kappa = 6$. The phase space points of the initial coherent state wavepackets are the same as in Fig. 3b), for which the averaged transmission probabilities are $|T_{\Psi_l}|^2 = |T_{\Psi_r}|^2 = 0.0125$. We have calculated $\overline{\Delta t}$ based on both exact quantum mechanical time evolution of the initial wavepackets and the classical Wigner prescription.

The numerical values of $\overline{\Delta t}$ for the two wavepacket propagation methods are virtually

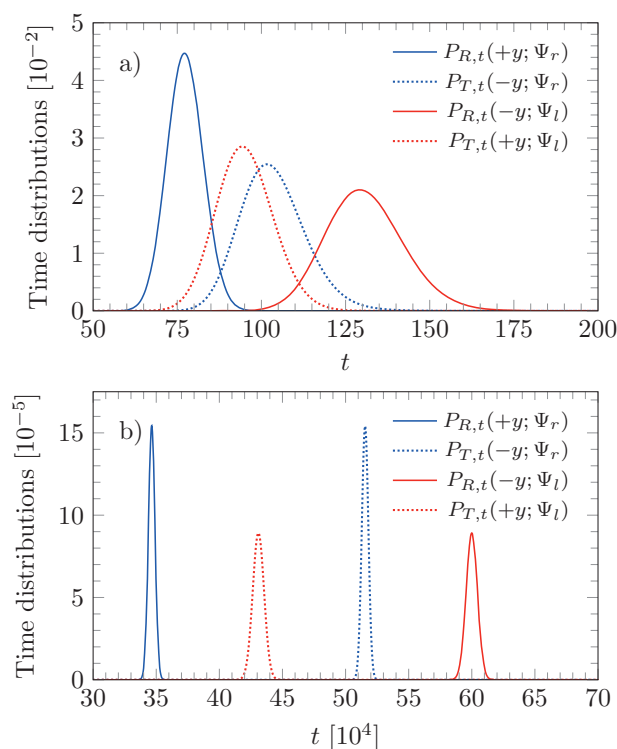


Figure 3: In panel a) transition path time probability distributions are calculated at $\pm y = \pm 75$ for the asymmetric Eckart barrier potential with $\kappa = 4$. The phase space center of the wavepackets incident from the left and the right are $(p_i, x_i) = (1, -50)$ and $(p_i, x_i) = (-\sqrt{3}, 50)$, respectively, and $\Gamma = 10^{-2}$. In panel b) transition path time probability distributions are calculated at $\pm y = \pm 4 \cdot 10^5$ with $\kappa = 6$. The phase space center of the wavepackets incident from the left and the right are $(p_i, x_i) = (1, -2 \cdot 10^5)$ and $(p_i, x_i) = (-\sqrt{3}, 2 \cdot 10^5)$, respectively, and $\Gamma = 5 \cdot 10^{-8}$. In all cases the initial conditions correspond to the energy $E = 3/2$.

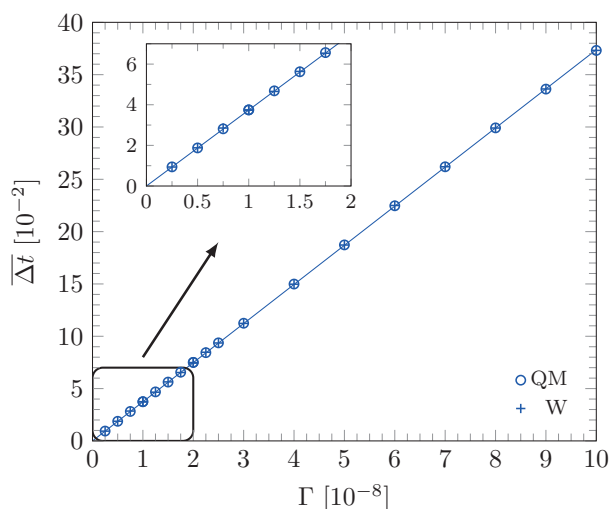


Figure 4: The overall flight time difference $\overline{\Delta t}$ as a function of the wavepacket width parameter Γ . The phase space center of the wavepackets incident from the left and right are $(p_i, x_i) = (1, -2 \cdot 10^5)$ and $(p_i, x_i) = (-\sqrt{3}, 2 \cdot 10^5)$, respectively, corresponding to the energy $E = 3/2$. The transition path time probability distributions are evaluated at $\pm y = \pm 4 \cdot 10^5$, and the parameter κ is 6. The calculation of $\overline{\Delta t}$ is based on both the exact quantum mechanical (QM) and classical Wigner (W) propagation methods.

identical based on visual inspection of Fig. 4, attesting to the accuracy of the approximate Wigner method, which has been noticed also in our previous studies of barrier scattering.^{22,24} In order to quantify the difference between the propagation methods, we calculate the average relative difference between the two sets of data points

$$\Omega = \sqrt{\frac{1}{N} \sum_{j=1}^N \frac{[\overline{\Delta t}_{\text{QM}}(\Gamma_j) - \overline{\Delta t}_{\text{W}}(\Gamma_j)]^2}{\overline{\Delta t}_{\text{QM}}^2(\Gamma_j)}}, \quad (3.1)$$

and we get $\Omega = 2 \cdot 10^{-3}$ in the specific case of Fig. 4. We can conclude that the classical Wigner propagation method is highly accurate for calculating the overall flight time difference based on transition path time probability distributions far from the barrier region. The Wigner method would obviously perform poorly close to the barrier, where interference effects observed in the exact quantum propagation are lost in the Wigner prescription due to the real-valued probabilities carried by the non-interacting classical trajectories.

The two sets of data points fit very well to the linear function $\overline{\Delta t}(\Gamma) = \alpha\Gamma$, where

$\alpha \approx 3.7 \cdot 10^6$ and the estimated standard errors are ~ 700 and ~ 6300 for the numerically exact quantum data and the classical Wigner data, respectively, i.e., the estimated standard errors are roughly three orders of magnitude smaller than the proportionality constant α . The apparent accuracy of the linear fit to the data points strongly indicates that the tunneling flight time $t_{\text{TFT}} = \lim_{\Gamma \rightarrow 0^+} \overline{\Delta t}(\Gamma)$, which is obtained from extrapolation of the data points, is zero for wavepacket transmission through the asymmetric Eckart barrier potential. The tunneling flight time obtained from the approximate Wigner prescription will be identically zero, since the classical trajectories are assumed to tunnel instantaneously through the barrier. The excellent agreement of the classical Wigner prescription with the numerically exact quantum dynamics further justifies our assertion that the tunneling flight time vanishes.

We have repeated these calculations for the five values of κ shown in Fig. 1. The corresponding transmission probability $|T_\Psi|^2$ (for the chosen range of Γ , the transmission probabilities are identical within five significant digits, i.e., $|T_\Psi|^2 = |T_{\Psi_l}|^2 = |T_{\Psi_r}|^2$), the proportionality constants $\alpha_{\text{QM/W}}$ and estimated standard errors based on the exact quantum data (QM) and the classical Wigner data (W), and the average relative difference Ω as given in Eq. (3.1) are all reported in Table 1. In Appendix B, we derive an approximate expression for the slope of the mean time difference as a function of Γ which is based on considering only the most probable trajectory for the transmitted wavefunction and the incident momentum for the reflected part. Moreover, the time differences are estimated as if the motion was that of a free particle only. The estimated slope based on this simple analytical expression is denoted by α_{th} in Table 1.

As is evident from the Table, we cover a large range of transmission probabilities (three orders of magnitude), and in all these cases the linear function $\overline{\Delta t}(\Gamma) = \alpha\Gamma$ is a very accurate representation of the numerical data points obtained from both quantum and classical Wigner propagation, e.g., the estimated standard errors are all roughly three orders of magnitude smaller than the corresponding proportionality constants. Furthermore, the average

Table 1: Numerical data for the linear dependence of $\overline{\Delta t}$ on Γ

κ	$ T_\Psi ^2$	$\alpha_{\text{QM}}/10^3$	$\alpha_{\text{W}}/10^3$	Ω	$\alpha_{th}/10^3$
4	0.4193	3098.5 ± 3.1	3099.4 ± 4.8	$9.4 \cdot 10^{-5}$	2208.2
5	0.0792	3643.4 ± 1.3	3646.9 ± 5.5	$3.4 \cdot 10^{-3}$	3501.4
6	0.0125	3749.3 ± 0.7	3744.7 ± 6.3	$2.0 \cdot 10^{-3}$	3755.1
7	0.0022	3765.5 ± 0.8	3772.6 ± 4.3	$1.3 \cdot 10^{-3}$	3794.3
8	0.0004	3768.1 ± 1.0	3760.7 ± 5.3	$1.2 \cdot 10^{-2}$	3801.0

relative difference Ω between the two data sets is at most $\sim 10^{-2}$, which indicates that the classical Wigner prescription is a very accurate approximation to the quantum dynamics independently of the transmission probability, and that the tunneling flight time vanishes for transmission through an asymmetric barrier.

We also note that in the deep tunneling region the simple free particle based slope α_{th} is in quite good agreement with the numerically exact quantum slopes, providing another piece of evidence that the central difference between the reflected and transmitted times in the deep tunneling regime is associated with momentum filtering, but not directly with a finite tunneling time.

4 Concluding remarks

We have studied the tunneling flight time for wavepacket transmission through the asymmetric Eckart barrier potential, which serves as a model of the potential along the reaction coordinate of a chemical system. For a large range of transmission probabilities the tunneling flight time vanishes. In view of the same conclusion reached previously for tunneling through a parabolic barrier,¹⁸ a symmetric Eckart barrier,²² and a symmetric square barrier,²² it seems that this is a general result for tunneling through one-dimensional time-independent barriers.

The flight time differences obtained from the exact quantum mechanical time evolution and the classical Wigner prescription are virtually identical, which underlines not only the ac-

curacy of the approximate Wigner method but also strengthens the claim that the tunneling flight time vanishes. In the Wigner prescription, the tunneling flight time is identically zero. The excellent agreement between the numerically exact and classical Wigner wavepacket propagation methods suggests that quantum coherence is not very important when it comes to pure tunneling. The same quality of agreement was also obtained previously for tunneling through symmetric barriers²² so that a general picture emerges from these studies, tunneling is the same irrespective of the specifics of the shape of the potential, as long as it is a single barrier separating reactants and products.

The vanishing tunneling time raises again the question of violation of special relativity. Of course, the theory used here is non-relativistic quantum mechanics and so in principle there is no contradiction if special relativity is violated. However, it would be worrisome, since it would imply that any reaction which involves tunneling is not treated correctly within non-relativistic quantum mechanics. As argued though in Ref. 22 (there is an error in Eq. (8) of Ref. 22, the correct version is given in Ref. 26) there is nothing to worry about. In any time-of-flight experiment one would never measure such a speed. Due to the constraints on the initial wavepacket imposed by the need to expose the pure tunneling time, i.e., prevention of unwanted effects such as above barrier scattering or leakage of the incident wavepacket into the product region, the time gained by the tunneling process is negligible compared to the overall flight time. In any time-of-flight experiment one will always measure a time which is characteristic of the incident momentum and the large distance between source and detector. The effect of the vanishing tunneling time is at best, very small.

The conclusion of a vanishing tunneling time has thus been fairly well established for one dimensional scattering. Although we do not expect major surprises when considering tunneling in multidimensional systems, it is necessary to repeat the computations presented in this paper for multidimensional systems, to understand if and where the added dimensions affect the tunneling flight time. Similarly, and especially in view of the attosecond photo-ionization experiments^{7,9-11} it is necessary to study the tunneling flight time in the presence

of an external time dependent field. A related question is the connection between tunneling flight times and tunneling times as determined by Larmor clock experiments.^{27–32}

Appendix A: Long time limit of the transition path time distribution

In this Appendix we provide an alternative proof to that given in Ref. 23 that the long time limit of the transition path time distribution goes as t^{-3} . The derivation presented below shows that this long time limit is a consequence of threshold quantum reflection, i.e., it is a purely quantum effect.

We assume a potential which becomes constant asymptotically, with energy $V_{-\infty} \geq 0$ when $x \rightarrow -\infty$ and $V_{\infty} = 0$ when $x \rightarrow \infty$. Noting the completeness of the scattering eigenstates we may write down the time evolved state formally exactly as in Eq. (2.21). The asymptotic form of the scattering eigenfunctions has been given in Eqs. (2.17)–(2.20).

We consider then the overlap $\langle x | \Psi(t) \rangle$ for the case that $x \rightarrow \infty$, x_i is large and negative, and the central momentum of the initial state is positive:

$$\langle x | \Psi(t) \rangle = \frac{1}{2\pi} \int_0^{\infty} dk T(k) \exp \left(\frac{i}{\hbar} \left(\sqrt{\hbar^2 k^2 + 2MV_{-\infty}} x - \frac{\hbar^2 k^2 t}{2M} \right) \right) \times \quad (\text{A.1})$$

$$\int_{-\infty}^{\infty} dx' \frac{1}{\sqrt{2\pi\hbar}} [\exp(-ikx') + R^*(k) \exp(ikx')] \langle x' | \Psi \rangle. \quad (\text{A.2})$$

Due to the localization of the incident coherent state about the positive momentum p_i and negative value of x_i it is sufficient to use only the positive momentum contribution of $\langle k^+ | x' \rangle$.

It is then a matter of Gaussian integration over the variable x' to find:

$$\langle x|\Psi(t)\rangle = \left(\frac{1}{\pi\hbar^2\Gamma}\right)^{1/4} \frac{1}{2\pi} \int_0^\infty dk T(k) \times \quad (\text{A.3})$$

$$\exp\left(\frac{i}{\hbar}x\sqrt{\hbar^2k^2 + 2MV_{-\infty}} - \frac{p_i^2}{2\hbar^2\Gamma} - \frac{k^2}{2\Gamma}\left(1 + i\frac{\hbar\Gamma t}{M}\right)\right) \times \quad (\text{A.4})$$

$$\left[\exp\left(-ik\left(x_i + i\frac{p_i}{\hbar\Gamma}\right)\right) + R^*(k) \exp\left(ik\left(x_i + i\frac{p_i}{\hbar\Gamma}\right)\right)\right]. \quad (\text{A.5})$$

Introducing the variable

$$\rho = \hbar k \sqrt{\left(1 + i\frac{\hbar\Gamma t}{M}\right)} \quad (\text{A.6})$$

allows us to rewrite the time dependent overlap as:

$$\langle x|\Psi(t)\rangle = \left(\frac{1}{\pi\hbar^2\Gamma}\right)^{1/4} \frac{1}{2\pi\hbar\sqrt{\left(1 + i\frac{\hbar\Gamma t}{M}\right)}} \int_0^\infty d\rho T\left(\frac{\rho}{\sqrt{\left(1 + i\frac{\hbar\Gamma t}{M}\right)}}\right) \times \quad (\text{A.7})$$

$$\exp\left(\frac{i}{\hbar}x\sqrt{\frac{\rho^2}{\left(1 + i\frac{\hbar\Gamma t}{M}\right)}} + 2MV_{-\infty} - \frac{p_i^2}{2\hbar^2\Gamma} - \frac{\rho^2}{2\hbar^2\Gamma}\right) \times \quad (\text{A.8})$$

$$\left[\exp\left(-\frac{i\rho\left(x_i + i\frac{p_i}{\hbar\Gamma}\right)}{\hbar\sqrt{\left(1 + i\frac{\hbar\Gamma t}{M}\right)}}\right) + R^*\left(\frac{\rho}{\sqrt{\left(1 + i\frac{\hbar\Gamma t}{M}\right)}}\right) \exp\left(\frac{i\rho\left(x_i + i\frac{p_i}{\hbar\Gamma}\right)}{\hbar\sqrt{\left(1 + i\frac{\hbar\Gamma t}{M}\right)}}\right)\right] \quad (\text{A.9})$$

At long times this becomes

$$\lim_{t \rightarrow \infty} \langle x|\Psi(t)\rangle \simeq \left(\frac{1}{\pi\hbar^2\Gamma}\right)^{1/4} \frac{1}{2\pi\hbar} \frac{1}{\sqrt{\left(1 + i\frac{\hbar\Gamma t}{M}\right)}} \exp\left(\frac{i}{\hbar}x\sqrt{2MV_{-\infty}} - \frac{p_i^2}{2\hbar^2\Gamma}\right) \times \quad (\text{A.10})$$

$$\int_0^\infty d\rho \exp\left(-\frac{\rho^2}{2\hbar^2\Gamma}\right) T\left(\frac{\rho}{\sqrt{i\frac{\hbar\Gamma t}{M}}}\right) \left[1 + R^*\left(\frac{\rho}{\sqrt{i\frac{\hbar\Gamma t}{M}}}\right)\right]. \quad (\text{A.11})$$

One then notes that the argument of the transmission and reflection amplitudes is very small in this limit. Quantum threshold reflection implies that

$$\lim_{p \rightarrow 0} R(p) \sim -1 + \frac{\eta}{2}p \quad (\text{A.12})$$

and

$$\lim_{p \rightarrow 0} T(p) \sim \gamma p \quad (\text{A.13})$$

so that in this long time limit

$$\lim_{t \rightarrow \infty} \langle x | \Psi(t) \rangle \simeq \left(\frac{1}{\pi \hbar^2 \Gamma} \right)^{1/4} \frac{\eta^* \gamma M \sqrt{M}}{4\pi \hbar^2 \Gamma \sqrt{i \hbar \Gamma}} \frac{1}{t^{3/2}} \times \quad (\text{A.14})$$

$$\exp \left(\frac{i}{\hbar} x \sqrt{2MV_{-\infty}} - \frac{p_i^2}{2\hbar^2 \Gamma} \right) \int_0^\infty d\rho \exp \left(-\frac{\rho^2}{2\hbar^2 \Gamma} \right) \rho^2 \quad (\text{A.15})$$

$$= \left(\frac{1}{\pi \Gamma} \right)^{1/4} \frac{\eta^* \gamma M \sqrt{M}}{2\sqrt{2\pi i}} \exp \left(\frac{i}{\hbar} x \sqrt{2MV_{-\infty}} - \frac{p_i^2}{2\hbar^2 \Gamma} \right) \frac{1}{t^{3/2}} \quad (\text{A.16})$$

demonstrating the generality of the long time limit which goes as $t^{-3/2}$ and that this limit comes expressly from the quantum threshold structure of the reflection and transmission amplitudes.

Appendix B: Theoretical estimate for the slope of the mean time difference

In this Appendix we derive an explicit expression and estimate for the slope of the mean time difference between the reflected and transmitted wavepackets in the case of asymmetric scattering. We assume that the wavepacket is incident from either the left or the right, centered at $-x$ or x ($x > 0$) respectively. The mean time is then estimated at $\pm y$ ($y > 0$) with y sufficiently larger than x so that initially the amplitude of the wavepacket at $\pm y$ is negligible. The incident wavepacket from the left (right) has momentum $\hbar k'$ ($|\hbar k|$) such that $\hbar^2 k'^2 = \hbar^2 k^2 - 2MV_{-\infty}$. In other words the potential at $x \rightarrow \infty$ vanishes and is lower by $V_{-\infty}$ relative to the potential at $x \rightarrow -\infty$. The incident wavepacket is a coherent state characterized by width Γ and incident from the left (right) with absolute value of the momentum p'_i ($|p_i|$) and the energy is measured relative to the potential at $x \rightarrow \infty$, that is $E = p^2/(2M)$.

Following the notation of the Eckart potential as in Eq. (2.13) we denote

$$\Omega^2 = -\omega^2 = \frac{1}{4} \left(\frac{8MV_{-\infty}b^2\kappa}{\hbar^2} - 1 \right). \quad (\text{B.1})$$

Assuming that Ω is real one can readily show, using the relations

$$|\Gamma(iy)|^2 = \frac{\pi}{y \sinh(\pi y)} \quad (\text{B.2})$$

$$|\Gamma(1 + iy)|^2 = \frac{\pi y}{\sinh(\pi y)} \quad (\text{B.3})$$

$$\left| \Gamma\left(\frac{1}{2} + iy\right) \right|^2 = \frac{\pi}{\cosh(\pi y)} \quad (\text{B.4})$$

that the absolute value of the reflection and transmission amplitudes given in Eqs. (2.14) and (2.15) simplifies to:

$$|R(k', k)|^2 = \frac{[\cosh^2(\pi\Omega) + \sinh^2(\pi b[k - k'])]}{[\cosh^2(\pi\Omega) + \sinh^2(\pi b[k + k'])]} \quad (\text{B.5})$$

and

$$|T(k', k)|^2 = \frac{\sinh(2\pi b k) \sinh(2\pi b k')}{[\cosh^2(\pi\Omega) + \sinh^2(\pi b[k + k'])]}. \quad (\text{B.6})$$

With these preliminaries we note that when incident from the left the distribution of incident momenta $P(k')$ which are transmitted may be written as

$$P(p' = \hbar k') = N_{p'} \exp \left[-\frac{(p' - p'_i)^2}{\hbar^2 \Gamma} + \ln \left(|T(p' = \hbar k', p = \hbar k(k'))|^2 \right) \right]. \quad (\text{B.7})$$

Noting that the width parameter is small, the most probable incident momentum is then readily found to be

$$\begin{aligned} p'_{mp} &= p'_i + \frac{\hbar^2 \Gamma}{2} \left[\frac{d}{dp'} \ln \left(|T(p' = \hbar k', p = \hbar k(k'))|^2 \right) \right]_{p'=p'_i} \\ &\equiv p'_i + \Delta p'_i. \end{aligned} \quad (\text{B.8})$$

Similarly, the most probable incident momentum which is transmitted when coming from the right is

$$\begin{aligned} p_{mp} &= p_i + \frac{\hbar^2 \Gamma}{2} \left[\frac{d}{dp} \ln \left(|T(p' = \hbar k'(p), p = \hbar k)|^2 \right) \right]_{p=p_i} \\ &\equiv p_i + \Delta p_i. \end{aligned} \quad (\text{B.9})$$

Although one could also take into consideration that reflected particles are slowed down since the reflection probability decreases with increasing energy in the tunneling regime, this decrease is exponentially small and so may be neglected.

We are now in a position to estimate the time difference between reflected and transmitted particles due to the filtering effect. Consider first particles incident from the left at the most probable incident momentum. The time it takes them to move the distance x needed to reach the origin, estimated as free particle motion, is Mx/p'_{mp} . They then continue to move from the origin to the point y but now their momentum, dictated by energy conservation is $\sqrt{p'^2_{mp} + 2MV_{-\infty}}$ so that the time needed for the transmitted portion is $My/\sqrt{p'^2_{mp} + 2MV_{-\infty}}$. For the reflected particles, the time needed to reach the barrier and then travel to $-y$ is $M(y+x)/p'_i$.

Similar estimates for the particle incident from the right are that for the transmitted particle the time taken to move the distance x needed to reach the origin is Mx/p_{mp} . The time for moving from the origin to the point y (the momentum is dictated by energy conservation) is $My/\sqrt{p^2_{mp} - 2MV_{-\infty}}$. For the reflected particles, the time needed to reach the barrier and then travel to $-y$ is $M(y+x)/p_i$. We thus have that the mean time difference between the reflected and the transmitted times is:

$$\bar{\Delta}t(\Gamma) = \frac{M}{2} \left[\frac{(y+x)}{p'_i} - \frac{x}{p'_{mp}} - \frac{y}{\sqrt{p'^2_{mp} + 2MV_{-\infty}}} + \frac{(y+x)}{p_i} - \frac{x}{p_{mp}} - \frac{y}{\sqrt{p^2_{mp} - 2MV_{-\infty}}} \right]. \quad (\text{B.10})$$

Since the incremental momenta Δp_i and $\Delta p'_i$ are small as compared to p_i and p'_i and are

linear with respect to the width parameter Γ we may consider only the first order correction to the mean time difference as a function of Γ to find:

$$\bar{\Delta}t(\Gamma) = \frac{M}{2} \left[\frac{x}{p_i^2} \Delta p_i' + \frac{yp_i'}{p_i^3} \Delta p_i' + \frac{x}{p_i^2} \Delta p_i + \frac{yp_i}{p_i^3} \Delta p_i \right]. \quad (\text{B.11})$$

Using this expression one obtains the theoretical values of the slope, denoted as α_{th} in Table 1.

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